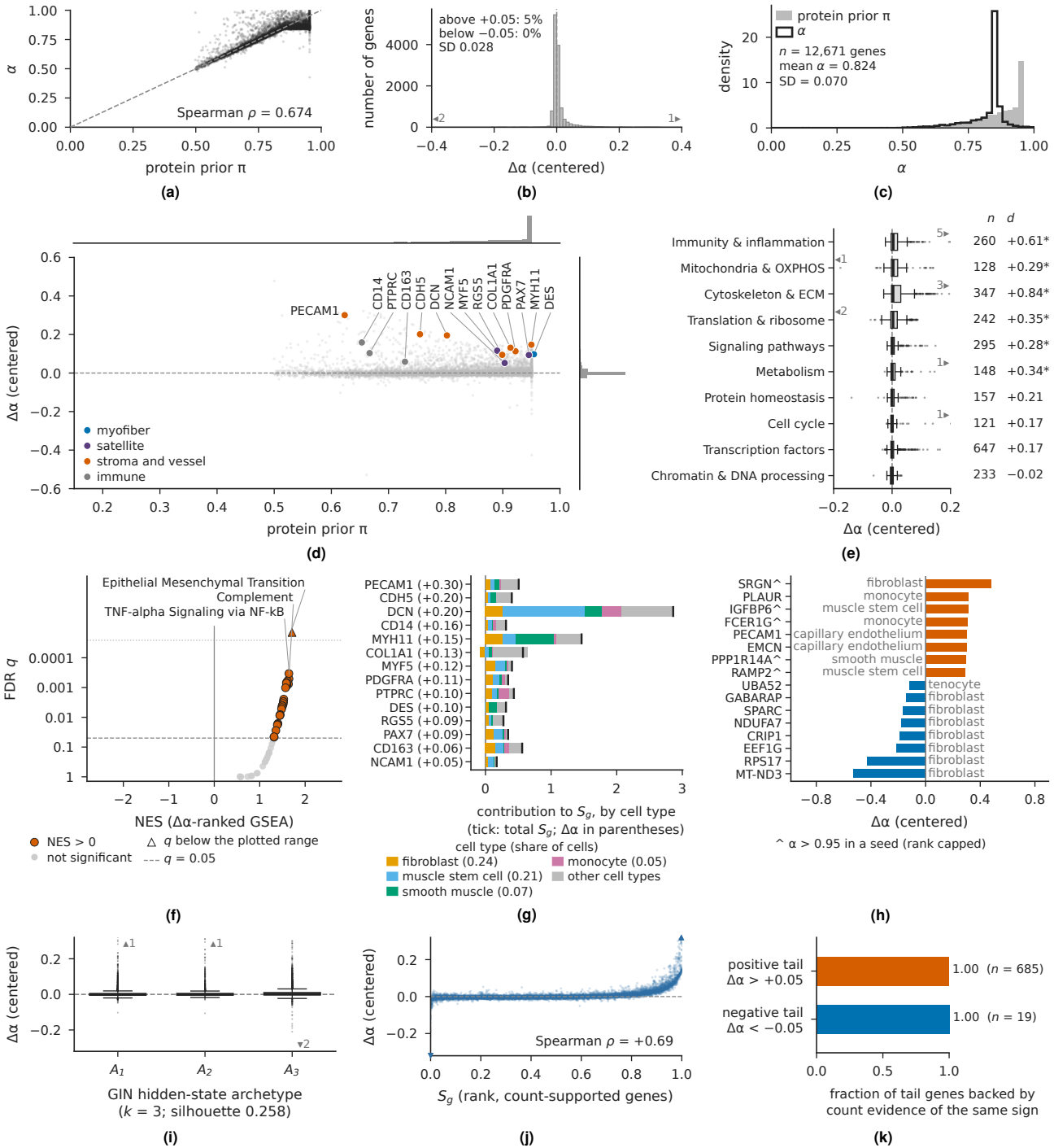

Spearman ρ = 0.674
protein prior π
α
(a)
above +0.05: 5%
below −0.05: 0%
SD 0.028
number of genes
Δα (centered)
(b)
protein prior π
α
n = 12,671 genes
mean α = 0.824
SD = 0.070
density
(c)
PECAM1
CD14
PTPRC
CD163
CDH5
DCN
NCAM1
MYF5
RGS5
COL1A1
PDGFRA
PAX7
MYH11
DES
myofiber
satellite
stroma and vessel
immune
Δα (centered)
protein prior π
(d)
n d
Immunity & inflammation 260 +0.61*
Mitochondria & OXPHOS 128 +0.29*
Cytoskeleton & ECM 347 +0.84*
Translation & ribosome 242 +0.35*
Signaling pathways 295 +0.28*
Metabolism 148 +0.34*
Protein homeostasis 157 +0.21
Cell cycle 121 +0.17
Transcription factors 647 +0.17
Chromatin & DNA processing 233 −0.02
Δα (centered)
(e)
Epithelial Mesenchymal Transition
Complement
TNF-alpha Signaling via NF-kB
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q below the plotted range
q = 0.05
(f)
PECAM1 (+0.30)
CDH5 (+0.20)
DCN (+0.20)
CD14 (+0.16)
MYH11 (+0.15)
COL1A1 (+0.13)
MYF5 (+0.12)
PDGFRA (+0.11)
PTPRC (+0.10)
DES (+0.10)
RGS5 (+0.09)
PAX7 (+0.09)
CD163 (+0.06)
NCAM1 (+0.05)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
fibroblast (0.24)
monocyte (0.05)
muscle stem cell (0.21)
other cell types
smooth muscle (0.07)
(g)
SRGN^ fibroblast
PLAUR monocyte
IGFBP6^ muscle stem cell
FCER1G^ monocyte
PECAM1 capillary endothelium
EMCN capillary endothelium
PPP1R14A^ smooth muscle
RAMP2^ muscle stem cell
UBA52 tenocyte
GABARAP fibroblast
SPARC fibroblast
NDUFA7 fibroblast
CRIP1 fibroblast
EEF1G fibroblast
RPS17 fibroblast
MT-ND3 fibroblast
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
GIN hidden-state archetype
(k = 3; silhouette 0.258)
(i)
Spearman ρ = +0.69
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
1.00 (n = 685)
negative tail
Δα < −0.05
1.00 (n = 19)
fraction of tail genes backed by
count evidence of the same sign
(k)